Proposed Plan of Study

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Whose Contribution Counts? AI Mediation and Collective Authorship in Digital Improvisation

In collaborative creative tools, a generative model reinterprets what participants contribute: one person's phrase becomes a sound or a moving form in a shared scene, and its outputs may become more visible than the contributions that prompted them. My doctoral research asks how participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, carries creative authority, and is credited in a collective work. I will examine whether technical attribution, the system's record of who supplied what, and social recognition among collaborators come apart. The question concerns communication and cultural participation: whose interpretations gain authority, and how authorship is negotiated when a machine takes part in creative exchange.

My objects of study are small-group digital improvisation workshops with adult improvising artists and creative practitioners, and the shared audiovisual environment I will build for them. Participants contribute short phrases. An existing language model proposes transformations into light, form and sound, and contributors or groups can accept, revise, reject or undo them. Anyone can take part without using AI.

I plan to use the program's research-creation option. The environment and the improvised works made with it form the creative component, and developing its rules and workshop scores is part of the inquiry. The methods are:

- scoping interviews with improvisers and facilitators;
- co-design with practitioners;
- repeated workshops (provisionally three small groups, four sessions each) comparing contributor-led, group-led and human-mediated arrangements;
- reflection interviews in which participants review video of their own sessions.

I will analyse episodes by linking talk, action and system records, and analyse interviews thematically. The framework is Lucy Suchman's account of how agency is figured at the human–machine interface, complemented by improvisation studies' attention to listening and response.

The project fits Technology in Practice, with Media & Culture as a minor. I hope to work with Dr. Rebecca Caines, whose research on improvisation as a research methodology, socially responsive creative technologies, and generative AI in research-creation fits this approach.

My background may appear distant from communication and culture, so I want to explain how it prepares me. I hold a Bachelor of Architecture from Tabriz Islamic Art University and a Master of Landscape Architecture from Istanbul Technical University. There I studied how people perceive virtual environments, work published at eCAADe 2023. I am completing a thesis-based master's degree in Interactive Arts and Technology at Simon Fraser University, where I built and evaluated multiplayer Unity systems, including ambient cues that let remote teammates perceive one another's heart-rate state without emotion labels; two manuscripts from

this work are under revision. Participants made acceptance of that display conditional on controlling who could see it, which led me to ask who controls how a shared system interprets people. I have conducted ethics-approved mixed-methods studies, worked as a teaching assistant in animation and interactive media, and served on SFU's Senate Graduate Studies Committee.